Network Proximity of Oncogenic Proteins

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1. Introduction

In this paper, we investigate the network proximity of cancer driver genes within the human Protein-Protein Interaction (PPI) network, motivated by the intuition that oncogenic proteins cluster near one another in the network. We evaluate whether known oncogenic proteins are significantly closer to one another than expected by chance and predict 30 novel cancer driver protein candidates. To achieve this, two distinct metrics were employed: Shortest Path Distance via Dijkstra's algorithm and Random Walk with Restarts (RWR). We validate our findings through a degree-preserving randomization procedure based on edge swapping.

2. Methods and Analysis

2.1 Pre-processing

2.1.1 Data

We analyzed two datasets provided by Dr. Borislav Hristov, bioinformatic researcher and professor at CalPoly, San Luis Obispo.

- *data/interacting_proteins.txt*: Pairs of proteins empirically found to bind to one another or participate jointly in the formation of larger molecular complexes within the cell's cytoplasm. This dataset comes from the The National Center for Biotechnology Information (NCBI) at Bethesda, MD, curated database of human interacting proteins which is the result of decades of laboratory work by numerous scientists.
- *data/onco_genes.txt*: Seven of the 18 confirmed cancer driver genes identified in The Cancer Genome Atlas (TCGA) kidney cancer study, used as the seed set throughout our experiments.

2.1.2 Largest Connected Component (LCC)

The PPI network is stored as an n * n, unweighted, undirected adjacency matrix A, where n is the number of unique proteins, A[i][j] = 1 indicates a validated interaction, and A[i][j] = 0 indicates no interaction. Alongside the matrix A, we maintain a protein index dictionary mapping each protein name to its corresponding row and column in A, allowing us to efficiently look up any protein by name. Because the graph contains disconnected components, we restrict our analysis to the Largest Connected Component (LCC), extract the corresponding submatrix, re-index the protein dictionary, and confirm that all seven seed oncogenes are present before proceeding.

2.2 Evaluation Criteria and Hypothesis Testing

Proteins sharing a biological function tend to interact with many of the same partners, placing them close to one another in the PPI. Cancer driver proteins, which share pathways for cell growth, division, and

survival, are expected to be proximal in the network. We quantify this with two metrics and evaluate each by comparing the oncogene score against a background of 1,000 randomly sampled protein sets of the same size, then compute a p-value as the fraction of random sets that matched or exceeded the oncogene score. We use a significance threshold of alpha = 0.10 throughout this experiment.

2.2.1 Shortest Path (Dijkstra)

For each pair of proteins in the oncogene set, we compute the minimum number of edges separating them in the LCC using Dijkstra's algorithm, then average over all pairs. A lower average indicates the proteins are more tightly clustered. We build a background by drawing 1,000 random sets of 7 non-oncogenic proteins from the LCC, computing the same average for each, and constructing the distribution. The p-value is the fraction of random non-oncogenic sets whose average was less than or equal to the oncogene average score.

2.2.2 Random Walk with Restarts (RWR)

We also apply Random Walk with Restarts (RWR), an algorithm that captures the full topology of the network rather than just the shortest path between proteins. The proximity score for the oncogene set is the average stationary probability assigned to the oncogene nodes at convergence, where convergence is defined as the Euclidean magnitude of the difference between consecutive probability vectors falling below 1e-10, or a maximum of 1,000 iterations as a safety net. We build a background distribution and compute a p-value similarly to the shortest path analysis, with the difference that the p-value is the fraction of random non-oncogenic sets whose score **matched or exceeded** the oncogene score. The full derivation and intuition for RWR are covered in the Appendix.

2.3 Graph Randomization

To confirm that the proximity score depends on the actual biological interactions and not merely on the degree distribution of nodes, we randomize the graph using degree-preserving edge swapping. In each swap, two edges (A, B) and (C, D) are replaced by (A, D) and (C, B), or (A, C) and (B, D) given neither new edge already exists. This preserves every node's degree while destroying the biologically meaningful wiring structure.

For sufficient randomization, we used 20|E| swaps, where E is the set of edges. On the randomized graph, we repeated both the shortest path and RWR analyses with 1,000 random background sets each, evaluating whether the oncogene proximity holds after randomization.

3. Results

3.1 Dijkstra vs. RWR

On the original PPI network, RWR produced a p-value of 0.067, significant at alpha = 0.10, confirming that the oncogene set is in closer network proximity than expected by chance. Dijkstra, however, produced a p-value of 0.416 and failed to reject the null hypothesis that oncogenes are no closer than a random set of proteins. It is important to note that this is expected because the shortest path captures only the single most direct route between two proteins and misses the topological structure of the network.

On the randomized graph, the RWR p-value increased to 0.682 and Dijkstra's to 0.145 (neither significant). Once the actual protein interactions were destroyed, both metrics produced scores indistinguishable from random, confirming that the proximity observed in the original network is a product of true biological topology rather than the degree distribution of the nodes.

Metric	Randomized Graph	p-value	Significant (alpha=0.1)
RWR	False	0.067	True
RWR	True	0.682	False
Dijkstra	False	0.416	False
Dijkstra	True	0.145	False

Table 1: Network Proximity Scores on Original and Randomized PPI

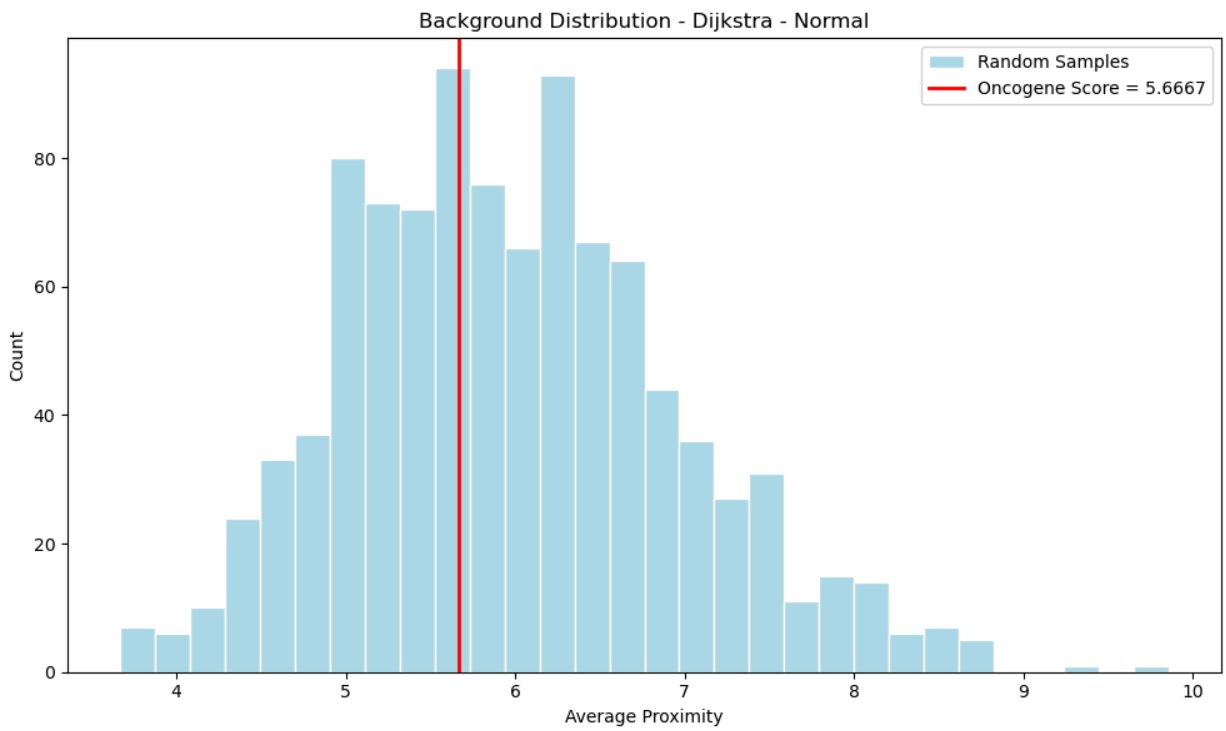


Figure 1: The Background distribution of Disktra is approximately normal, and the oncogenes is not sufficiently left-shifted to reach statical significance

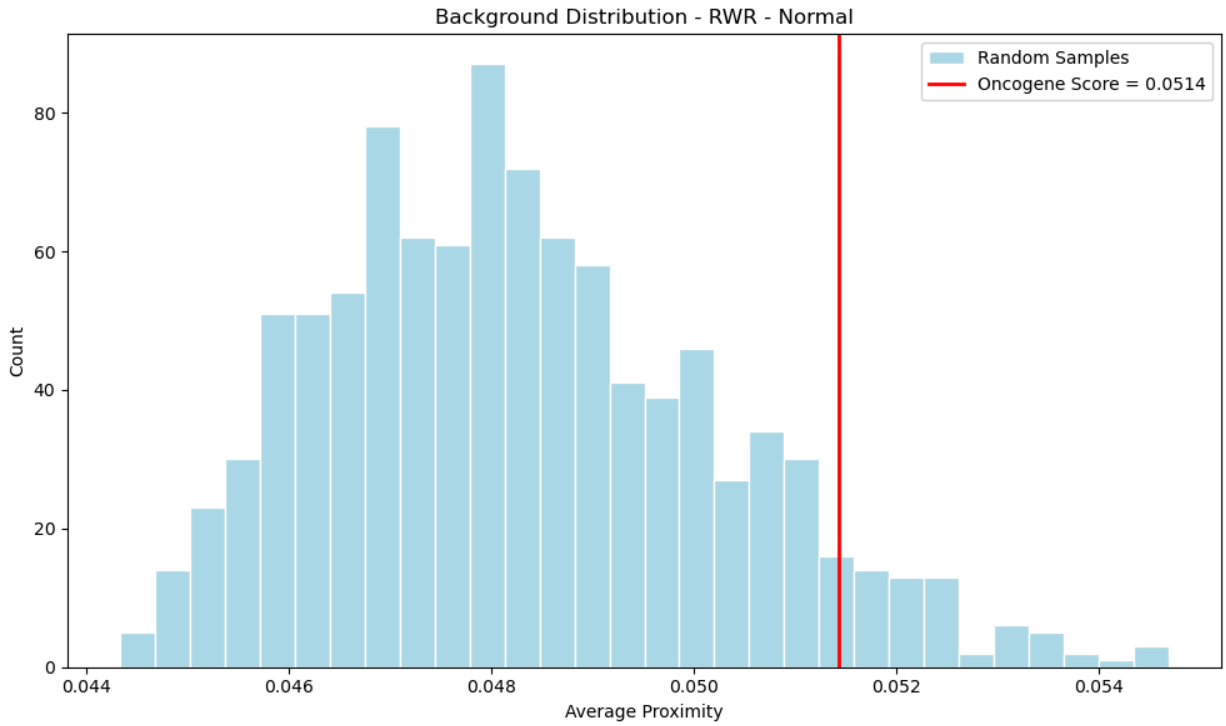


Figure 2: The background distribution of RWR is approximately normal, and the oncogene is sufficiently right-shifted to reach statistical significance.

3.2 Predicting Novel Oncogenic Genes

Using RWR, we rank all proteins in the LCC by their stationary probability when starting with the seven known oncogenes. The higher the stationary probability, the more connected a protein is to the known cancer drivers through the network, which makes it a stronger candidate for a novel oncogenic role. We excluded the seed proteins and stored the top 30 as predicted novel cancer driver proteins at *Sahand_Aslani_onco_predictions.txt* and *Sahand_Aslani_onco_predictions.png*.

Appendix

Random Walk with Restarts (RWR)

This section develops a thorough understanding of the Random Walk with Restarts algorithm (RWR), covering the mathematical intuition, the effect of edge weights and the restart parameter, and the rank correlation analysis between number of degrees and stationary probabilities.

Algorithm and Derivation

We start by building a transition matrix B from the adjacency matrix A, where each row of A is divided by the node's degree so that the probabilities of moving to any neighbor sum to 1. The probability of going from node i to node j is stored in B[i][j], so to get the probability of arriving at j from any node, we need to look down the column of j across all possible i's.

We then initialize two vectors: (1) start_prob holds the initial probability mass, split uniformly over the seed nodes and zero everywhere else, and it never changes throughout the algorithm, serving as the teleportation mechanism. (2) walk_prob holds the result of one step of pure walking, computed as B transposed times the current probability vector, representing where the walk would land if it never teleported.

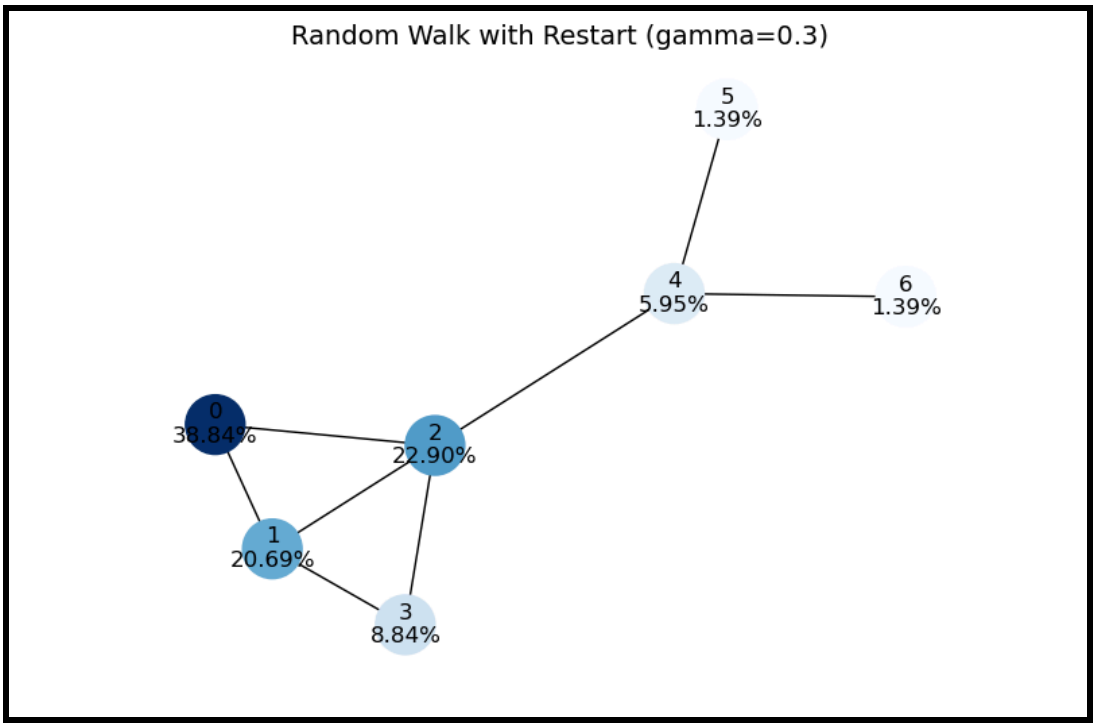
At each iteration, two **mutually exclusive** and **exhaustive** events can happen: the walk either teleports back to the seed with probability gamma, or moves to a neighbor with probability (1 - gamma). Since these are the only two possibilities, by the law of total probability the chance of being at node j next is:

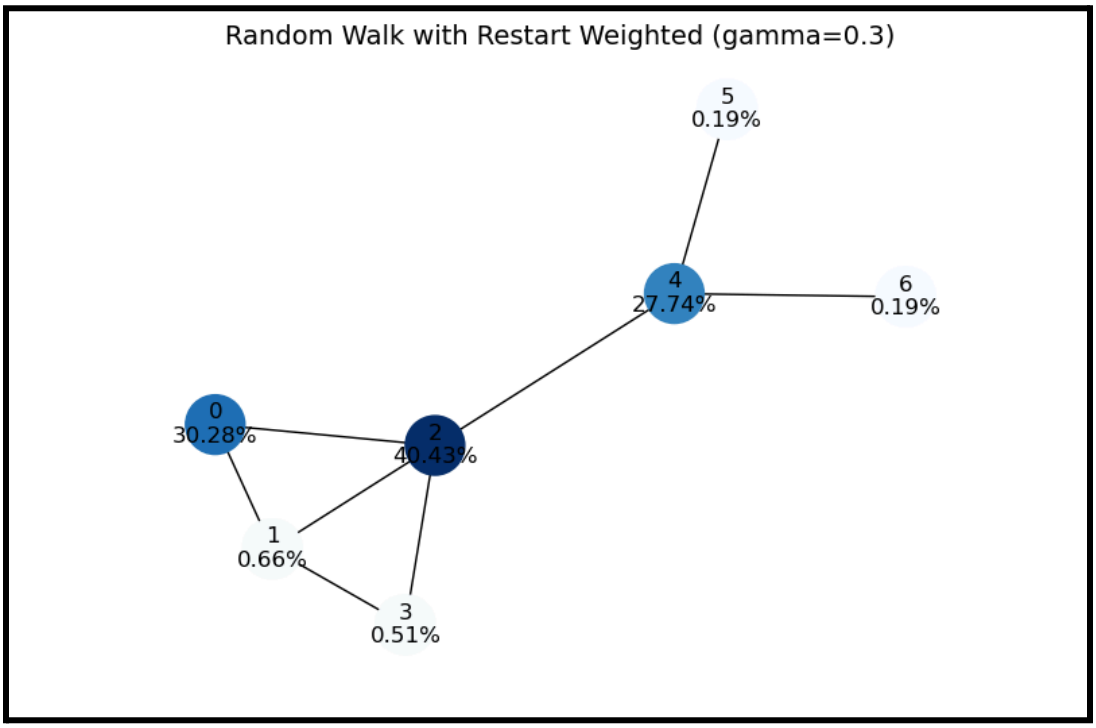
P(at j) = P(teleport) * P(at j|teleport) + P(walk) * P(at j |walk)
= gamma * start_prob[j] + (1 - gamma) * walk_pro[j]

We repeat this update until the Euclidean magnitude of the difference between consecutive probability vectors falls below the threshold, at which point the distribution has converged and we return the final vector. We also set a maximum of 1,000 iterations as a safety net, halting execution if convergence is not reached.

Effect of Weights on the edges

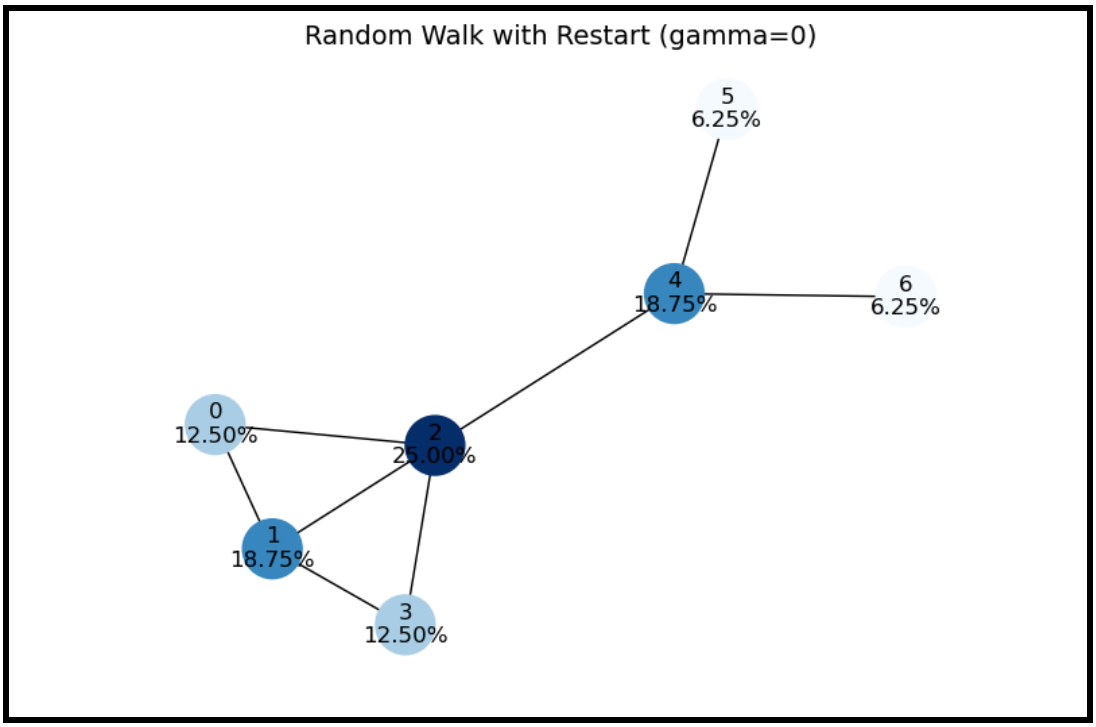
With uniform weights, nodes 0, 1, 2, and 3 capture most of the probability mass given their tight interconnection and proximity to the seed (node 0). When we widen the pipe between nodes 2 and 4 by a factor of 100 and narrow the pipe between 0 and 1 by the same, more flow rushes toward the outer cluster of 4, 5, and 6 while the seed cluster loses some of its dominance.

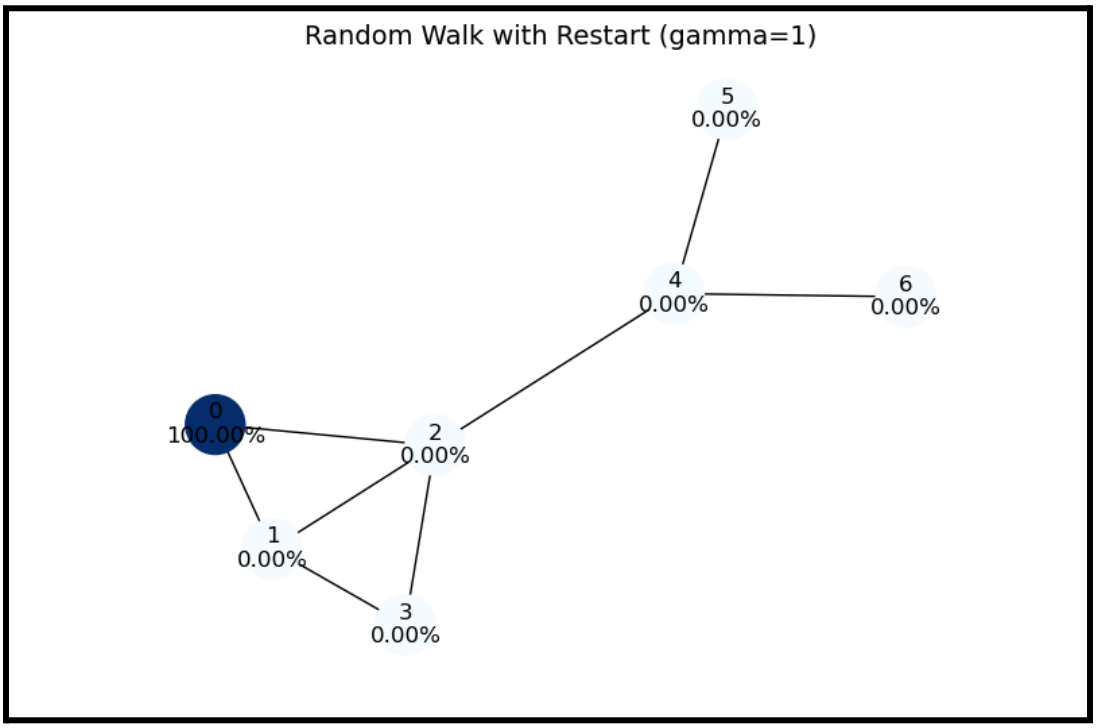




Effect of Restart Parameter

At gamma = 0.0 the walk explores freely, and node 2 overtakes node 0 as the highest probability node because its three edges make it the most connected in the graph. At gamma = 1.0, the walk never leaves the seed, so node 0 holds all probability mass and everything else is exactly zero.





Spearman Correlation between Degrees and Stationary Probabilities

At gamma = 0.0, the stationary distribution mirrors node degree almost perfectly, confirming a statistically significant relationship driven entirely by network topology. At gamma = 1.0, the correlation

collapses to near zero and is not significant, as degree becomes irrelevant when the walk never leaves the seed. At gamma = 0.3 the correlation is moderate and not significant, reflecting a walk that is still anchored to the seed while exploring the network.

Gamma (Restart Prob)	Spearman Correlation	P-values	Interpretation
0.0	0.99	0.0	~ Perfect Correlation
1.0	-0.10	0.82	~ No Correlation
0.3	0.63	0.13	Moderate

Implementation Details

Analysis implemented in Python using:

- *numpy*: Adjacency matrix creation, matrix operations, and numerical computation
- *scipy*: LCC extraction, Dijkstra Shortest Path and Spearman Rank Correlation
- *networkx*: Graph creation, visualization, double edge swapping randomization and degree count
- *matplotlib*: Rendering

Running *main.py* generates the output files in *./outputs* directory:

- *Sahand_Aslani_onco_predictions.png*
- *Sahand_Aslani_onco_predictions.txt*
- *dijkstra_background_Normal.png*
- *rwr_background_Normal.png*

Running *rwr_intuition.py* generates the output files in *./outputs* directory:

- *rwr_simulation.png*
- *rwr_weighted_simulation.png*
- *rwr_simulation_gamma_0.png*
- *rwr_simulation_gamma_1.png*

The complete analysis was run on an Apple M4 Pro. The *rwr_intuition* pipeline completed in a few seconds. The *main* pipeline took approximately 30 minutes. Both pipelines are initiated by running the respective script, assuming all dependencies are installed.